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Use of the Columbia-Suicide Severity Rating Scale Screener in a General Hospital Setting

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ABSTRACT

Specific differences between interviewer-rated and self-report suicide risk assessments are unknown. This descriptive cross-sectional study examined the acceptance of the Columbia-Suicide Severity Rating Scale (C-SSRS) screener, conducted by nurse-interviewers versus self-reporting, and identified the factors contributing to refusal in each mode. We reviewed the charts of inpatients at a general hospital that included the C-SSRS screener at the time of admission. C-SSRS screening by a nurse-interviewer and self-reported mode were included. Two modes and the acceptors and refusers were compared using descriptive statistics. Multi-variate logistic regression was performed to determine associations between the studied factors and suicide risk. The study included 398 patients. The acceptance of the nurse-interview and self-report modes was 93.8% and 74.7%, respectively. Refusers for the former were more likely to be male ($p = 0.029$) and from the surgical departments ($p < 0.001$). Refusers for the latter tended to be older ($p < 0.001$). Compared to the nurse-interview group, the self-report group was more likely to report a high risk of suicide (odds ratio = 6.508, $p = 0.041$). Each mode of the C-SSRS screener, when used in general hospital inpatients, has its own advantages and disadvantages. Healthcare providers should select the mode optimal for their institutional environment after comprehensively considering the patients' characteristics and available hospital resources.

1 | Introduction

Annually, the number of deaths resulting from suicide exceeds 700 000 individuals (Korea Foundation for Suicide Prevention 2023). Physical illness, as well as mental disorders, increases the likelihood of death by suicide (Viilo et al. 2005). Concurrent physical and psychiatric problems are also commonly noticed among inpatients who died by suicide (Suominen et al. 2002). Therefore, early detection of mental disorders and suicide risks is recommended for patients with physical illness who are being treated in a general hospital setting. Previous research shows that patients with suicidal ideation in non-psychiatric wards have less communication about their suicidal ideation with healthcare providers and tend to attempt suicide after their admission sooner than those in psychiatric wards (Cheng et al. 2009). Screening to identify suicidal ideation among inpatients with physical illness by their primary healthcare providers could be useful (Idenfors et al. 2019).

2 | Background

Suicide risk evaluation for inpatients may be easily overlooked because there are time constraints due to the heavy workload

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among general hospital personnel, and the management of medical or surgical problems for which patients are admitted is prioritised. Consequently, screening scales for assessing suicide risk have been widely used in clinical settings (Thom et al. 2020). The Columbia-Suicide Severity Rating Scale (C-SSRS) (Posner et al. 2011) is one of the most used instruments for evaluating suicidal ideation and behaviours. Its screener version (The Columbia Lighthouse Project 2023) has also been used in clinical and research settings (Dunn et al. 2021; Omolewa and Tribble 2018; Walters 2020). Each mode of the screener, whether self-report or interviewer-rated, appears to have its pros and cons. For example, a self-reported assessment has the advantage of saving time for healthcare personnel, but the interviewer-rated assessment is regarded as more caring. However, it is not known which mode is preferred by patients and is superior in detecting high-risk individuals. In a large institution such as a general hospital, once a system has been set up, changes including switching modes are perceived as burdensome, costly and complex (e.g., establishing a new protocol, training personnel and updating contents in electronic devices). Therefore, it is important to consider various factors in depth before implementing one mode over another. To our knowledge, no studies have directly compared potential differences between the two screening modes.

Hence, we did a chart review of general hospital inpatients who were screened via either the self-report or the interviewer-rated mode of the C-SSRS screener. In this exploratory study, we examined the acceptance of each mode, identified refusal contributors of each mode and determined the differences between the two modes in detecting individuals who are at a high risk of suicide. We aimed to provide insights into the factors to consider before adopting either mode in the general hospital setting.

3 | Methods

3.1 | Design

We conducted a descriptive cross-sectional study.

3.2 | Participants and Setting

3.2.1 | Programme Development and Purpose

The Asan suicide risk assessment and management programme (ASAP) at the Asan Medical Center, Seoul, Korea, was developed by the Department of Psychiatry and implemented by the Clinical Performance Improvement Team and the Department of Nursing. The purpose of the programme was to ensure that individuals with a high risk of suicide are detected promptly and referred to consultation-liaison psychiatrists for timely management.

3.2.2 | Phases I and II

The ASAP was conducted in two phases. Phase I was conducted between 5 June 2020 and 21 June 2020 at two oncology wards with a high prevalence of distressed patients with cancer (Zabora et al. 2001). Phase II was performed between 6 December and 13 December 2020. Phase II was an extension of Phase I and included a variety of medical and surgical departments (Table S1) in eight wards, one of which was the same as in Phase I. The Department of Psychiatry did not participate in the programme; thus, no mental health nurses participated in the study.

3.2.3 | Patient Assessment and Data Collection

Each ward independently implemented either mode of the C-SSRS screener—the nurse-interview or the self-report mode. Specific modes were assigned according to the wards; all the patients in each ward were allocated to either the self-report or nurse-interview group. Consecutive patients who were admitted to the participating wards were assessed using the C-SSRS screener at the time of their admission. Due to the explorative nature of the study, the sample sizes for each category were not predetermined. To gather sufficient information, instead, we reviewed the medical charts of all the patients who had completed the screening during the phases.

3.3 | Variables

The Korean version of the C-SSRS screener was developed by the ASAP group.

i. Forward translation: During the translation process, the lead author (C.H.K.P.), a suicidologist-psychiatrist who is bilingual in English and Korean, initially translated the version while also similarly translating the C-SSRS rater version, which had been validated in Korean (Jang et al. 2014).

ii. Reconciliation: The questionnaire was reviewed for cultural relevance by our study team members, including nurses and psychiatrists, to ensure that all items were contextually appropriate and understandable in the Korean cultural setting. The team reached a consensus on the draft of the Korean translation of the C-SSRS screener, confirming that it accurately conveyed both the literal meaning and conceptual intent of the original English version.

iii. Back translation: A bilingual psychiatrist fluent in both English and Korean (S.C.) and unfamiliar with the original English version of the C-SSRS screener conducted an independent back translation of the Korean version into English.

iv. Back translation review and finalisation: The study team compared the back-translated version with the original instrument to verify both its literal and conceptual equivalence. Adjustments were made where necessary to align with local norms and practices. Each mode of the C-SSRS screener is composed of six questions, answered with a yes or no. Respondents are classified as no, low, moderate or high risk of suicide. Patients are considered to be at high risk of suicide if they have had any thoughts or intentions of killing themselves, or have developed

some ideas of how to kill themselves, or had an intention to implement their plan in the past 1 month, or, if they have done, begun to do or prepared to do anything to take their own lives in the last 3 months. Demographic and clinical variables including age, sex, medical/surgical departments, whether each patient accepted or refused the C-SSRS screener and the C-SSRS score were collected.

3.4 | Data Analysis

Characteristics between the interviewer-rated and the self-report modes were compared for the pre- and post-screening group. The pre-screening group is one that was originally provided with a specific mode, and the post-screening group is one that actually completed the specific mode. For the age distribution in each group, normality was tested using the Kolmogorov–Smirnov test. The independent *t*-test was used for age in case that normal distribution was confirmed; otherwise, the Mann–Whitney *U* test was conducted. The chi-squared test or Fisher's exact test was performed for categorical variables. We defined acceptance of a screening mode (%) as the number of acceptors divided by the number of patients who were provided with that screening mode, multiplied by 100. Acceptors and refusers were compared using descriptive statistics. Next, differences in age, sex, medical/surgical department and screening mode among the suicide risk groups were compared with a one-way analysis of variance or a chi-squared test. To examine associations of the factors with each level of suicide risk, multi-variate logistic regression was performed using 'no risk' as a reference category. Characteristics between the acceptors and the refusers of psychiatric referrals were also compared. A two-tailed *p* value < 0.05 was considered statistically significant. All statistical analyses were conducted using SPSS for Windows version 24.0 for PC (IBM Corp, Armonk, NY, USA).

3.5 | Ethical Considerations

The Institutional Review Board of Asan Medical Center approved the current study (2021-0790). Informed consent was waived because this study was based on a retrospective chart review.

4 | Results

During both phases, 400 patients were admitted, 398 of whom completed either mode of the screening.

4.1 | Comparisons Between the Nurse-Interview and Self-Report Groups

Demographic and clinical variables were compared between the nurse-interview and self-report groups in Table 1. Only for the age distribution in the post-screening self-report group, normality was confirmed (Table S1). Both pre- and post-screening self-report groups were more likely to be from medical departments ($\chi^2 = 96.970$, $p < 0.001$ for the former and $\chi^2 = 94.981$, $p < 0.001$ for the latter). The patients in the post-screening group were younger ($t = 3.424$, $p = 0.001$).

4.2 | Acceptance of Screening Modes and Characteristics of the Acceptors and Refusers

Demographic and clinical variables were compared between the acceptors and refusers for each screening mode in Table 2. The acceptance of the nurse-interview and self-report modes was 93.8% and 74.7%, respectively (Table 3). For the age distribution in each group except the acceptors of the nurse-interview mode, normality was confirmed (Table S1). The nurse-interview mode was more likely to be acceptable ($\chi^2 = 29.461$, odds ratio [OR] = 5.130, $p < 0.001$). For the nurse-interview mode, refusers were more likely to be male ($\chi^2 = 4.764$, $p = 0.029$) and from surgical departments ($p < 0.001$). As there was only one patient from a medical department who rejected the interview mode, we performed logistic regression without the department variable. Sex remained statistically significant after adjusting for age (OR = 5.264, 95% confidence interval [CI] = 1.144–24.228, $p = 0.033$). For the self-report mode, refusers were more likely to be older ($t = -4.922$, $p < 0.001$). Statistical significance remained after adjusting for sex and department (OR = 1.057, 95% CI = 1.026–1.089, $p < 0.001$).

4.3 | Risk Groups and Suicide Risk Factors

Of those who completed the screening, 349 (87.7%), 34 (8.5%), 8 (2.0%) and 7 (1.8%) showed no, low, moderate and high risk, respectively. No statistical differences in the studied variables among the risk groups were found (Table 4).

Compared to the nurse-interview group, the self-report group was associated with an increased probability of reporting a high risk of suicide (OR = 6.508, 95% CI = 1.075–39.384, $p = 0.041$), compared with no risk, after adjusting for all other covariates (Table 4). Compared to those from medical departments, patients from surgical departments were more likely to be at moderate risk of suicide than at no risk (OR = 0.127, 95% CI = 0.026–0.624, $p = 0.011$) after adjusting for all other variables.

Of the seven high-risk patients, three (42.86%) refused psychiatric referral. There were no statistically significant between-group differences in age, sex or department (Table S3).

5 | Discussion

In this study, we examined the acceptance of the nurse-interview and self-report mode of the C-SSRS screener, factors associated with refusal of each mode and variables related to a high risk of suicide. The acceptance of the nurse-interview mode was significantly higher than that of the self-report mode. Males and surgical patients were identified as refusers of the nurse-interview mode. Older age was associated with refusing the self-report mode. Moreover, the patients in the self-report group were more likely to report a high risk of suicide than those from the nurse-interview group.

TABLE 1 | Comparisons between the nurse-interview group and the self-report group.

Variables	Total (%)	Nurse-interview (%)	Self-report (%)	Test statistic (χ^2 or t)	p
Pre-screening					
Patients	400 (100.0)	242 (60.5)	158 (39.5)	—	—
Age, *M* (SD), years	58.4 (14.8)	59.0 (14.7)	57.4 (14.9)	1.056	0.246
Sex					
Female	176 (44.0)	97 (40.1)	79 (50.0)	3.816	0.051
Male	224 (56.0)	145 (59.9)	79 (50.0)		
Department					
Medical[a]	285 (71.3)	216 (89.3)	69 (43.7)	96.970	**< 0.001**
Surgical[b]	115 (28.8)	26 (10.7)	89 (56.3)		
Post-screening					
Patients	398 (100.0)	265 (66.6)	133 (33.4)	—	—
Age, *M* (SD), years	58.4 (14.8)	60.2 (13.9)	54.7 (15.9)	3.424	**0.001**
Sex					
Female	175 (44.0)	115 (43.4)	60 (45.1)	0.106	0.745
Male	223 (56.0)	150 (56.6)	73 (54.9)		
Department					
Medical	283 (71.1)	230 (86.8)	53 (39.8)	94.981	**< 0.001**
Surgical	115 (28.9)	35 (13.2)	80 (60.2)		

Note: Significant findings at $p < 0.050$ are in bold.
Abbreviations: *M*, mean; SD, standard deviation.
[a]Includes internal medicine (oncology, gastroenterology, nephrology and endocrinology) and dermatology.
[b]Includes surgery (hepatobiliary and pancreatic surgery, liver transplantation surgery, acute care surgery, vascular surgery, colon and rectal surgery, breast surgery and endocrine surgery), ophthalmology, dentistry, urology and neurosurgery.

As more than half of our patients were from the department of oncology (Table S1), and some patients from other departments were also cancer patients, we compared the self-report acceptance from our study (74.7%) with that of previous studies in which cancer patients were assessed using a self-report screening tool for psychiatric problems such as depression and anxiety. Our results were comparable to previous studies (77% (Park et al. 2021) and 586/740 (79.18%) (Roth et al. 1998)). The slight difference in acceptance between the former and the latter might have stemmed from differences in the screening purpose (suicide risk vs. general psychiatric issues), group characteristics (cancer plus some non-cancer patients vs. cancer patients only) and the type of questionnaire. However, the nurse-interview mode showed a high percent of acceptance (93.8%). To our knowledge, this is the first study to examine the acceptance of a screening interview conducted by a rater and to investigate the acceptance of scales for suicide risk screening. Both modes were deemed to be generally acceptable.

Overall, the self-report mode had over five times more refusals than the nurse-interview mode (Table 3). As the reasons for refusal are unavailable, we can only speculate with caution that some patients may have felt more uncomfortable in marking their suicidal symptoms on paper by themselves than in expressing them verbally to a nurse. A previous study reported that, when self-reporting, depressive symptoms may be underreported (Hunt et al. 2003), which could be related to a stigma of psychiatric illness, known to be prevalent among Korean elderly patients (Park, Cho, et al. 2015). Even though the nurse-interview mode was more likely to be accepted, it cannot be concluded that it is a better measure in terms of acceptance because the variables differed between the acceptors and refusers of the different screening modes (Table 3). In either screening mode, being male, being surgical patients and older age were associated with refusals. These findings are in close agreement with those from previous studies (Walker et al. 2008), which reported difficulty in screening patients who are male and older. First, males and surgical patients were more likely to refuse the screening in the nurse-interview mode. Males tend to show stigmatising views on male depression and suicide (Oliffe et al. 2016), which can interfere with men seeking help and limit self-disclosure about depression or suicidal ideation (Livingston and Boyd 2010). Considering that demographic characteristics between the medical and surgical patients in this study did not differ ($t = 0.243$, $p = 0.808$ for age; $F = 0.285$, $p = 0.593$ for sex), and no significant differences exist in major psychiatric disorders between surgical and medical inpatients (Arolt et al. 1997), we speculate that nurse factors may have contributed to the difference in acceptance. Although there is no significant difference in the total workload between medical and surgical nurses (Al-Kandari and Thomas 2009), the former group tends to experience physical abuse from patients or visitors and receive complaints more

TABLE 2 | Comparisons between the acceptors and refusers for each screening mode.

Variables	Total	Acceptors (%)	Refusers (%)	Test statistic (χ^2 or t)	p
Nurse-interview					
N	242 (100.0)	227 (93.8)	15 (6.19)	—	—
Age, *M* (SD), years	59.0 (14.7)	59.4 (14.4)	53.5 (19.4)	1.508	0.333
Sex					
Female	97 (40.1)	95 (41.9)	2 (13.3)	4.764	**0.029**
Male	145 (59.9)	132 (58.1)	13 (86.7)		
Department					
Medical[a]	216 (89.3)	215 (94.7)	1 (6.7)	—[c]	**< 0.001**
Surgical[b]	26 (10.7)	12 (5.3)	14 (93.3)		
Self-report					
N	158 (100.0)	118 (74.7)	40 (25.3)	—	—
Age, *M* (SD), years	57.4 (14.9)	54.8 (15.6)	65.1 (9.6)	−4.922	**< 0.001**
Sex					
Female	79 (50.0)	58 (49.2)	21 (52.5)	0.134	0.714
Male	79 (50.0)	60 (50.8)	19 (47.5)		
Department					
Medical	69 (43.7)	52 (44.1)	17 (42.5)	0.030	0.863
Surgical	89 (56.3)	66 (55.9)	23 (57.5)		

Note: Significant findings at $p < 0.050$ are in bold.
Abbreviations: *M*, mean; SD, standard deviation.
[a]Includes internal medicine (oncology, gastroenterology, nephrology and endocrinology) and dermatology.
[b]Includes surgery (hepatobiliary and pancreatic surgery, liver transplantation surgery, acute care surgery, vascular surgery, colon and rectal surgery, breast surgery and endocrine surgery), ophthalmology, dentistry, urology and neurosurgery.
[c]Fisher's exact test.

TABLE 3 | Acceptance of the screening modes.

Variables	Acceptors (%) ($N = 242$)	Refusers (%) ($N = 158$)	Test statistic (χ^2)	p	OR
Nurse-interview	227 (93.8)	15 (6.2)	29.461	**< 0.001**	5.130
Self-report	118 (74.7)	40 (25.3)			

Note: Significant findings at $p < 0.050$ are in bold.
Abbreviation: OR, odds ratio.

often (Al-Kandari and Thomas 2008) and may be more used to providing defensive care. Given that suicide risk assessment addresses a highly private and delicate topic, and failure to identify high-risk patients could entail grave consequences and contribute to a patient's death, we speculate that medical nurses may have made an exceptional effort to complete the interview successfully. Surgical nurses have fewer opportunities to develop self-confidence and autonomy in nursing practice than medical nurses because of their different working environments (Adams and Bond 1997). Being used to a somewhat passive role, surgical nurses may be less likely to participate in decision-making (Hoffman et al. 2004), and therefore have difficulties in initiating, conducting and completing a skilful interview on a totally unfamiliar and highly sensitive topic, especially when patients showed an uncooperative attitude. Second, older age was associated with refusals in the self-report mode. Many studies have shown that older people view mental illness as a greater stigma (Jang et al. 2009; Park, Kim, et al. 2015). An increase in age is associated with a greater discriminatory stigma in those with depression and a history of suicide attempts (Park, Kim, et al. 2015). The perceived stigma of mental illness has also led to the underutilisation of mental health services (Park, Cho, et al. 2015). Together, the acceptance of each screening mode would depend on the characteristics of the population to which it is applied.

Regarding suicide risk detection, patients who used the self-report mode were more likely to report a high risk of suicide. As all other variables were adjusted for, this finding represents a difference between the modes themselves. First, the interviews

TABLE 4 | Association of the studied factors with each suicide risk[a].

	Low risk			Moderate risk			High risk		
Variables	**OR**	**95% CI**	***p***	**OR**	**95% CI**	***p***	**OR**	**95% CI**	***p***
Age	1.025	0.999–1.053	0.063	1.001	0.952–1.052	0.982	0.977	0.930–1.027	0.363
Sex									
Female	1.000			1.000			1.000		
Male	1.831	0.894–3.751	0.098	2.443	0.555–10.762	0.238	1.012	0.217–4.716	0.988
Department									
Medical[b]	1.000			1.000			1.000		
Surgical[c]	2.253	0.821–6.179	0.115	0.127	0.026–0.624	**0.011**	2.494	0.420–14.800	0.315
Screening mode									
Nurse-administered	1.000			1.000			1.000		
Self-reported	1.831	0.762–4.399	0.176	0.228	0.038–1.377	0.107	6.508	1.075–39.384	**0.041**

Note: Significant findings at $p < 0.050$ are in bold.
Abbreviations: CI, confidence interval; OR, odds ratio.
[a]Compared to no risk.
[b]Includes internal medicine (oncology, gastroenterology, nephrology and endocrinology) and dermatology.
[c]Includes surgery (hepatobiliary and pancreatic surgery, liver transplantation surgery, acute care surgery, vascular surgery, colon and rectal surgery, breast surgery and endocrine surgery), ophthalmology, dentistry, urology and neurosurgery.

were performed in an environment where privacy was not sufficiently protected as they were conducted at an open nurse station in the middle of a ward, or in inpatient rooms, most of which were shared by up to six patients. Inpatients are aware that their personal information could be overheard, which constrains disclosure of information pertinent to diagnosis and treatment (Malcolm 2005). Considering that a history of suicidal ideation and behaviour is highly personal, it is likely that the open interview space limited the disclosure of sensitive information during the screening. Second, stigma or misconception surrounding suicidal ideation or behaviour may have contributed to non-disclosure during an interview. A previous study showed that patients often withhold suicidal ideation from a screening interviewer because they regard suicidal ideation as shameful and an indication of weakness and worry that disclosure would result in involuntary hospitalisation or initiation of medication (Ganzini et al. 2013). Third, there may be psychological resistance to the screening interview process itself. Some suicidal ideators are reluctant to participate as they regard suicidal thoughts as private and the interview as intrusive. The unexpected and mechanical interviewing process may also be viewed as a violation of privacy by the interviewers (Ganzini et al. 2013). Whatever the reason, withholding information on suicidal ideation and behaviour hinders diagnosis and treatment. To improve the odds of identifying individuals at high risk of suicide in the nurse-interview mode, the above issues must be addressed.

Among the high-risk patients, almost 50% refused psychiatric consultation referral. Although replication using a large size of a study sample is necessary, these current findings suggest that there may be a number of potential referral refusers. The refusal rate in this study is comparable to those of previous studies on post-suicide attempt admissions. In these previous studies, most patients must have been at high risk of suicide, and refusers of psychiatric referrals accounted for 26%–68% (Elsaeiti et al. 2020; Lin et al. 2012; Belle 2016). Our finding is novel, and, given that these patients were at high risk of suicide, highly alarming. The result indicates that the nurses should anticipate a high referral refusal rate and be prepared to address it effectively. Namely, even for referral refusers, all routine measures to prevent suicide must immediately be implemented. The suicide management plan from the C-SSRS website (The Columbia Lighthouse Project 2021) introduces precaution procedures besides the formal psychiatric consultation, including identification of patients at high risk, restricting access to the means for one's suicidal plan, securing a safe environment such as special observation and making a psychiatric outpatient appointment within 3 days of discharge.

Each mode of the C-SSRS screener has its own advantages and disadvantages. Either mode can be chosen based on available resources, hospital capacity and patient characteristics. Considering the high refusal rate of the self-report mode for patients of older ages, the nurse-interview mode may be especially advantageous when used in institutions where elderly patients are hospitalised. The self-report mode was superior in detecting patients at a high risk of suicide, and this finding must also be seriously considered as older patients usually have a high risk for suicide (Shah et al. 2016). Moreover, without mental health professionals who can provide nurses with repetitive education and feedback on suicide risk assessment and management, the nurse-interview mode could not be implemented. However, the self-report mode could be adopted in hospitals where a majority of the patients are male or surgical patients since they are more likely to refuse the nurse-interview mode. The self-report mode may also be preferred in an environment where a heavy workload or high patient turnover is expected. Otherwise, increased

demands on patient care and adverse outcomes may result (Park et al. 2012). In an environment where access to a smartphone is available, applying the self-report mode can be favourably considered as the smartphone application has been shown to be useful for screening depression and suicide (Jang et al. 2017). These findings must be considered together with the aforementioned factors to assist healthcare providers in choosing the preferred mode for screening that is optimal to the hospital environment. If possible, a tailored approach for each patient group would be highly desirable.

Nurses are usually the first healthcare providers that newly admitted patients meet at a hospital. Thus, nurses have the first opportunity to make a timely assessment by screening for suicide risk and identify patients with suicidal ideation and advocate for intervention. Some research studies have shown that patients agree with the idea of being asked about their suicidal ideation by nurses at admission (Horowitz et al. 2013) or even at the emergency department (Ballard et al. 2012). A previous study showed that patients rate the interviewers as being more comfortable in asking questions about suicide than the interviewers rate themselves (Horowitz et al. 2013). A formal training programme to improve the nurses' ability to properly apply the screening tool and manage those identified with a high risk for suicide should be developed. Indeed, nurses need to recognise that assessing suicide risk is one of their vital responsibilities and the key to preventing suicides in the hospital setting (Bolster et al. 2015). Although the ASAP primarily involved non-mental health nurses, the importance of addressing the mental health needs of medical or surgical patients among them cannot be overlooked. Until consultation with mental health providers is available, they are likely to be 'front-line' in suicide prevention (Berlim et al. 2007). While non-mental health nurses may not typically primarily manage their patients' mental health issues, as healthcare professionals, they should possess the basic capability to assess and respond to suicide risk (Bolster et al. 2015) and can actually be equipped with fundamental knowledge about suicidal behaviour even after brief training (Berlim et al. 2007). Our study is related to a broader sense of mental health nursing because timely identification and effective management of mental health crises, such as suicidal tendencies, and vigilance with high-risk patients in medical or surgical wards are inevitable factors for the safe transfer to a psychiatric ward for more specialised mental health nursing and treatment. Additionally, regularly using assessment tools to identify suicide risk could enhance mental health awareness.

The current study has several limitations. First, because this study was conducted at a single hospital, there may be institutional bias. Generalisation of the findings through a multi-institutional study is required. Second, patient information regarding their history of psychiatric disorders and the severity of current diseases for which they were hospitalised was unavailable. Third, suicide is a culture-bound phenomenon. The degree to which disclosing a desire for suicide is socially acceptable can vary between cultures. As all the patients were Korean, applying these results to other ethnic groups should be conducted with caution. Fourth, the Korean version of the C-SSRS did not undergo the standard validation process, which may have led to reduced reliability and cultural bias.

6 | Conclusion

Both the nurse-interview and self-report modes of the C-SSRS screener, when used in general hospital inpatients, are acceptable, although the former showed higher acceptance. Older patients were more likely to refuse the self-report mode, whereas males and surgical patients tended to refuse the nurse-interview mode. Moreover, the self-report mode identified more patients with a high suicidal risk. A multi-centre study with a larger sample size is required to confirm the present findings.

7 | Relevance for Clinical Practice

Although nurses working in medical or surgical wards may not typically focus on managing their patients' mental health issues, they should have the basic skills to assess and respond to suicide risk before the possible transfer of high-risk patients to a psychiatric ward. To effectively screen inpatients, they should choose the mode of the C-SSRS screener that best fits their institutional environment, considering patient characteristics and available resources, because the nurse-interview and self-report modes of the C-SSRS screener have different practical acceptance characteristics.

Author Contributions

All authors listed meet the authorship criteria according to the latest guidelines of the International Committee of Medical Journal Editors and are in agreement with the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.